

Assignment – 3

Question – 1:

Code:

```
1 // If ages of Ram, Shyam, and Ajay are input through keyboard, then find the youngest of them all.
2
3 #include <stdio.h>
4 int main() {
5     int r, s, a, youngest;
6     printf("Enter The Ages of Ram, Shyam, and Ajay:\n");
7     scanf("%d %d %d", &r, &s, &a);
8     if(r == s && r == a) printf("All of them have the same age.\n");
9     else if (r==s && r<a && r!=a) printf("Ram and Shyam are the youngest.\n");
10    else if(r==a && r<s && r!=s) printf("Ram and Ajay are the youngest.\n");
11    else if(s==a && s<r && s!=r) printf("Shyam and Ajay are the youngest.\n");
12    else { (r > s ? (s > a ? (printf("Ajay is youngest.\n")) : (printf("Shyam is youngest.\n"))) : (r > a ? (printf("Ajay is youngest.\n")) : printf("Ram is youngest.\n"))); }
13 }
```

Output:

```
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q1
Enter The Ages of Ram, Shyam, and Ajay:
21
33
18
Ajay is youngest.
```

Question – 2:

Code:

```
1 // Write a program to determine whether a year entered is leap or not.
2
3 #include <stdio.h>
4 int main() {
5     int year;
6     printf("Enter a year: ");
7     scanf("%d", &year);
8     if(year % 4 == 0) {
9         if(year % 100 == 0) {
10            if(year % 400 == 0) printf("The year %d is a leap year.\n", year);
11            else printf("The year %d is not leap.\n", year);
12        } else printf("The year %d is a leap year.\n", year);
13    } else printf("The year %d is not leap.\n", year);
14 }
```

Output:

```
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q2
Enter a year: 2016
The year 2016 is a leap year.
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q2
Enter a year: 2100
The year 2100 is not leap.
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q2
Enter a year: 2400
The year 2400 is a leap year.
```

Question – 3:

Code:

```
1 // Take an integer input from the user and check if the number is divisible by 3.
2
3 #include <stdio.h>
4 int main() {
5     int num;
6     printf("Enter a number: ");
7     scanf("%d", &num);
8     if(num % 3 == 0) printf("The number %d is divisible by 3.\n", num);
9     else printf("The number %d is not divisible by 3.\n", num);
10 }
```

Output:

```
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q3
Enter a number: 37
The number 37 is not divisible by 3.
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q3
Enter a number: 36
The number 36 is divisible by 3.
```

Question – 4:

Code:

```
1 // Write a program to check wheter a triangle is valid or not if the three angles are input through keyboard.
2
3 #include <stdio.h>
4 int main() {
5     int an1, an2, an3;
6     printf("Enter the three angles of a triangle:\n");
7     scanf("%d %d %d", &an1, &an2, &an3);
8     if(an1 + an2 + an3 == 180) printf("The triangle is valid.\n");
9     else printf("The triangle is invalid.\n");
10 }
```

Output:

```
students@cse1abiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q4
Enter the three angles of a triangle:
90
45
45
The triangle is valid.
students@cse1abiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q4
Enter the three angles of a triangle:
60
61
62
The triangle is invalid.
```

Question – 5:

Code:

```
1 // If length of 3 sides of a triangle are input through keyboard, then check wheter the triangle is isosceles, equilateral, or scalene triangle.
2 #include <stdio.h>
3 int main() {
4     int an1, an2, an3, si1, si2, si3;
5     printf("Enter the three angles of a triangle:\n");
6     scanf("%d %d %d", &an1, &an2, &an3);
7     if(an1 + an2 + an3 == 180){
8         if(an1 == 0 || an2 == 0 || an3 == 0) printf("The triangle is invalid.\n");
9         else {
10             printf("Enter the three sides of a triangle:\n");
11             scanf("%d %d %d", &si1, &si2, &si3);
12             if(an1 == 90 || an2 == 90 || an3 == 90) printf("The triangle is right angled.\n");
13             (si1 == si2 ? (si1 == si3 ? (printf("The triangle is equilateral.\n")) : (printf("The triangle is isosceles.\n")))) : (printf("The triangle is scalene.\n"))));
14         }
15     }
16     else printf("The triangle is invalid.\n");
17 }
```

Output:

```
students@cse1abiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q5
Enter the three angles of a triangle:
90
90
0
The triangle is invalid.
students@cse1abiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q5
Enter the three angles of a triangle:
78
78
5
The triangle is invalid.
students@cse1abiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q5
Enter the three angles of a triangle:
90
45
45
Enter the three sides of a triangle:
44
34
56
The triangle is right angled.
The triangle is scalene.
```

Question – 6:

Code:

```
1 // Write a program using the ternary operator to check whether the character entered through keyboard is lowercase or not.
2
3 #include <stdio.h>
4 int main() {
5     char letter;
6     printf("Enter a letter: ");
7     scanf("%c", &letter);
8     if(letter >= 'a' && letter <= 'z') printf("The letter is lowercase.\n");
9     else if(letter >= 'A' && letter <= 'Z') printf("The letter is uppercase.\n");
10    else printf("The input is invalid.\n");
11 }
```

Output:

```
students@cseelabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q6
Enter a letter: f
The letter is lowercase.
students@cseelabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q6
Enter a letter: H
The letter is uppercase.
students@cseelabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_3$ ./q6
Enter a letter: 99
The input is invalid.
```